

Main references of the ContEst paper: methods, mathematics, and comparison with ContEst

ContEst supplementary note

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Abstract

This note collects the principal references cited in the ContEst paper, with an emphasis on the lines of work ContEst is compared against. Each section explains the referenced method, the mathematics it uses, its benefits, and—most importantly—how it relates to and differs from ContEst. A summary table at the end (Table 1) maps every reference class onto the design axes and gradient mechanism, mirroring the positioning table of the paper.

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1 What ContEst does, and the axes of comparison

ContEST (Control and Estimation Co-Design) poses actuation *and* sensing/estimation design in a single two-stage stochastic program. The design $\theta = (\theta_f, \theta_h)$ collects plant/actuator parameters

θ_f (entering the dynamics f) and sensing/estimator parameters θ_h (entering the output map h or the noise covariances), and is chosen to minimize

$$\min_{\theta \in \Theta} J_{\text{des}}(\theta) + J_{\text{stoch}}(p_0; \theta), \quad J_{\text{stoch}} = \min_{u_{0:N-1}} \frac{1}{N} \mathbb{E} \left\{ \sum_k \ell(x_k, u_k) + \ell^N(x_N) \right\}, \quad (1)$$

the second stage being the stochastic optimal-control (dual-control) value, expressed through the *information state* (filtering density) so the sensing operator enters the objective explicitly. The inner value is evaluated by a *convex* problem—two SDPs (or Riccati equations) in the LQG regime, a pair of game Riccati equations in the robust H_∞ regime, and a convex information-state MPC in the eKF-linearized nonlinear regime. The design gradient $\nabla_\theta J_{\text{stoch}}$ is obtained *exactly* from the dual variables the inner solver already returns, by the envelope theorem—no differentiation through the optimizer or its KKT system. Discrete sensor allocation is recovered by a sparse ℓ_1 relaxation.

We compare each reference on five axes: **(i)** is the plant/actuation designed? **(ii)** is the sensing/output map designed? **(iii)** is the *estimator itself* designed (e.g. a filter bandwidth, or the noise level $V(\theta)$), not merely a choice of sensors? **(iv)** is closed-loop control in the design objective? **(v)** how is the design gradient/derivative obtained? Nonlinear reach is a sixth, secondary axis.

2 LQG control and sensing co-design (nearest neighbor)

Tzoumas, Carlone, Pappas, Jadbabaie, “LQG Control and Sensing Co-Design,” IEEE TAC 2021 [1].

Method. This is the closest single reference to CONTEST. It jointly designs the sensing (which sensors to activate from a menu, under a sensor budget/cost) and the LQG controller over a finite horizon. Its central structural result is a *separation*: the optimal LQG control gains are independent of the sensor selection, so the co-design collapses to choosing the sensor set that minimizes the sensing-dependent part of the LQG cost, for which a polynomial-time algorithm with provable sub-optimality guarantees is given.

Mathematics. Finite-horizon LQG; two Riccati recursions—a control Riccati that does *not* see the sensors, and a Kalman filtering Riccati whose error covariance depends on the measurement model. The LQG cost is written as a sum of trace terms $\sum_t \text{tr}(\Theta_t \Sigma_t)$ of the filtering error covariances; sensor selection is a cardinality-constrained set-function problem, shown NP-hard in general, attacked by a greedy/relaxation algorithm with per-instance and worst-case bounds derived from the (approximate) supermodular structure.

Benefits. A principled, provably near-optimal joint treatment of sensing and LQG control; identifies the separation that makes the problem tractable; scalable greedy selection with guarantees.

Comparison with ContEst. Same spirit—design sensing *and* control together—but CONTEST differs on every axis. (1) *Discrete vs. continuous*: they pick a subset from a fixed sensor *menu*; CONTEST tunes *continuous* sensing/estimator parameters (precisions, placements on a continuous scale, filter bandwidths, noise levels $V(\theta)$), recovering the discrete case only when needed via an ℓ_1 relaxation. (2) *Actuation*: their plant/actuation is fixed; CONTEST co-designs θ_f on the same footing. (3) *Estimator design*: they select sensors but do not tune a dynamic estimator; CONTEST can design the estimator itself (e.g. the PLL bandwidth study). (4) *Where separation is used*: they lean on separation at the *design* level (menu selection given a fixed control law); CONTEST uses separation only *within* the inner solve and couples both axes through the shared θ , so the co-design does not separate. (5) *Gradient*: greedy/relaxation vs. an *exact* envelope gradient. (6) *Regime*: linear/LQG only vs. CONTEST’s nonlinear (eKF) and robust (H_∞) extensions. The paper’s PLL

study is a head-to-head against exactly the separation-style sequential pipeline that such menu methods induce.

3 Plant–controller coupling, CCD, and MDO

Fathy, Reyer, Papalambros, Ulsoy, “On the Coupling Between the Plant and Controller Optimization Problems,” *ACC 2001* [2]; *Herber & Allison*, “Nested and Simultaneous Solution Strategies...,” *JMD 2019* [3]; *Allison & Herber*, “Multidisciplinary Design Optimization of Dynamic Engineering Systems,” *AIAA J 2014* [4].

Method. This lineage founds *control co-design* (CCD). Fathy et al. prove that plant and controller optimizations are *coupled* and that sequential (plant-then-control) design is generally suboptimal. Herber & Allison systematize the two solution strategies—*nested* (an inner controller solve inside an outer plant search) and *simultaneous* (one optimization over both). Allison & Herber frame dynamic-system CCD as multidisciplinary design optimization (MDO).

Mathematics. Combined plant/control optimization with first-order (KKT) necessary conditions; the coupling is captured by cross-sensitivity terms $\partial^2 J / \partial x_p \partial x_c$ between plant x_p and control x_c variables—sequential design is stationary for the joint problem only when these vanish. Nested strategies form a bilevel program; simultaneous strategies a single NLP, typically via direct transcription of the dynamics.

Benefits. Establishes *why* co-design is necessary; provides a taxonomy and practical solution architectures for dynamic engineering systems.

Comparison with ContEst. CONTEST adopts this coupling thesis and extends it to the *estimation* axis: the sequential plant→sensing→estimator→control pipeline is a feasible but generally non-stationary point of the joint problem. Architecturally CONTEST is a *nested* (MDF-like) scheme, but its inner discipline is *convex* and its coupling derivative is supplied *exactly by duality* (the envelope gradient) rather than by a finite-difference sensitivity sweep or an adjoint through a nonconvex inner solve—the usual bottleneck of nested CCD. Crucially, the CCD literature keeps sensing/estimation fixed (full-state feedback or a fixed observer), so the output map never enters the design; CONTEST makes it a first-class design variable.

4 Structural and information-architecture LMI co-design

Grigoriadis, Zhu, Skelton, “Optimal Redesign of Linear Systems,” *ASME JDSMC 1996* [5]; *Li, de Oliveira, Skelton*, “Integrating Information Architecture and Control or Estimation Design,” *SICE JCMSI 2008* [6].

Method. The structural-control (Skelton) school casts integrated structure/actuator and control design as *linear matrix inequalities* (LMIs). Grigoriadis et al. alternate a control LMI with a plant/actuator-parameter LMI to redesign a linear system for better closed-loop covariance performance. Li et al. push this to the *information architecture*—sensor/actuator gains and precisions—treated as decision variables jointly with control or estimation.

Mathematics. Covariance (output-variance-constrained) control: express the stationary H_2 cost through a Lyapunov LMI in the closed-loop covariance Σ and a linearizing change of variables ($L = K\Sigma$), giving a convex SDP; alternate between the control-side and plant/architecture-side LMIs (block coordinate descent). This is exactly the ancestor of CONTEST’s control SDP.

Benefits. Each subproblem is convex; structural/architecture parameters enter tractably; mature LMI machinery and solvers apply.

Comparison with ContEst. CONTEST’s LQG control SDP is the Grigoriadis–Skelton output-variance-constrained program, and its estimation SDP is the dual. Three differences. (1) *Alternation vs. exact joint gradient*: they alternate LMIs (coordinate descent), whereas CONTEST reads the *exact* design gradient from the LMI dual by the envelope theorem and takes a joint quasi-Newton step. (2) *“and” vs. “or”*: Li et al. integrate architecture with control *or* estimation; CONTEST designs control *and* estimation together against one stochastic objective, coupling them through θ . (3) *Regime*: their LMIs are linear/ H_2 (or H_∞); CONTEST adds the nonlinear eKF-MPC surrogate and the fixed- γ game-Riccati robust route with the identical envelope gradient.

5 Sensor selection for estimation

Joshi & Boyd, “Sensor Selection via Convex Optimization,” IEEE TSP 2009 [7]; Tzoumas, Jadbabaie, Pappas, “Sensor Placement for Optimal Kalman Filtering,” ACC 2016 [8]; Zhang, Ayoub, Sundaram, “Sensor Selection for Kalman Filtering...,” Automatica 2017 [9].

Method. Select a subset of sensors that sharpens an estimation criterion, for a *fixed* plant and no control in the loop. Joshi & Boyd maximize a log-determinant information measure by a Boolean relaxation; Tzoumas et al. and Zhang et al. minimize Kalman filtering error by greedy algorithms with (near-)supermodular guarantees, and characterize when such guarantees hold.

Mathematics. Static case: Fisher information / error-covariance functionals, $\max_z \log \det (\sum_i z_i a_i a_i^\top)$ with $z \in \{0, 1\}^m$ relaxed to $[0, 1]^m$; convex, then rounded. Dynamic case: Kalman Riccati error covariance as a set function; submodularity/supermodularity of $\log \det$; greedy with $(1 - 1/e)$ -type bounds (and counterexamples for other metrics, per Zhang et al.).

Benefits. Convex or greedy, scalable to many candidate sensors, with optimality gaps or approximation guarantees; the canonical tools for pure sensor placement.

Comparison with ContEst. These are *estimation-only, open-loop, discrete*. They do not design actuation, do not tune a dynamic estimator, and do not put closed-loop control in the objective—the sensor set is optimized for a filtering metric, not for J_{stoch} . CONTEST embeds sensing inside the closed-loop stochastic control value, co-designs actuation, and returns *exact* design gradients; its estimation cost $\text{tr}(Q\Sigma^\varepsilon)$ is precisely an A-optimal-type criterion, so these methods are the open-loop special case of the sensing half of CONTEST. CONTEST’s ℓ_1 sensor allocation is the closed-loop analog of Joshi–Boyd’s Boolean relaxation.

6 Sparsity-promoting sensor/actuator selection

Dhingra, Jovanović, Luo, “An ADMM Algorithm for Optimal Sensor and Actuator Selection,” CDC 2014 [10]; Das & Bhattacharya, “Sparse Sensing Architecture for Kalman Filtering...,” 2017 [11]; Saraf, Bhattacharya, Skelton, “ H_2 Optimal Sensing Architecture...,” ACC 2017 [12].

Method. Promote *sparse* sensing/actuation by penalizing an ℓ_1 (or reweighted- ℓ_1) surrogate of the cardinality of the active set, minimizing an H_2 /Kalman performance measure; solve by ADMM or SDP, then threshold. Das et al. and Saraf et al. add guaranteed estimation-error bounds.

Mathematics. H_2 norm or Kalman error covariance as the smooth objective; $\lambda \|\cdot\|_1$ (or reweighted $\sum_i w_i |\cdot|_i$) as the sparsity penalty; ADMM splitting, or LMI/SDP with sparsity constraints; thresholding to a binary architecture.

Benefits. Convex-surrogate handling of the combinatorial selection; tunable sparsity–performance trade-off; error guarantees in the guaranteed-bound variants.

Comparison with ContEst. This family is the direct ancestor of CONTEST’s sensor-allocation mechanism (Section 7 of the paper): continuous gains $\alpha_j \geq 0$ scaling each sensor, penalized by $\lambda_s \|\alpha\|_1$ in J_{des} , with an epigraph lift for smoothness. The differences: they optimize for a *fixed* plant/control and a static or H_2 criterion; CONTEST places the same ℓ_1 in a *closed-loop* (possibly nonlinear) co-design that also moves actuation and estimator parameters, and keeps the envelope gradient untouched by the penalty.

7 Dual control and stochastic/adaptive MPC

Feldbaum, “Dual Control Theory,” 1960 [13]; *Bar-Shalom & Tse*, “Dual Effect, Certainty Equivalence, and Separation in Stochastic Control,” IEEE TAC 1974 [14]; *Mesbah*, “Stochastic MPC with Active Uncertainty Learning: A Survey on Dual Control,” ARC 2018 [15].

Method. Dual control recognizes that an input has a *dual effect*: it both regulates the state and probes to reduce uncertainty about it. Bar-Shalom & Tse formalize the dual effect, certainty equivalence (when probing is absent), and separation. Mesbah surveys how modern stochastic MPC approximates dual control, distinguishing *active* (probing) from *passive* learning.

Mathematics. Stochastic dynamic programming over the *information state*—the conditional density $p(x_k | \mathcal{Z}_k)$; the value function depends on this density, and certainty equivalence holds iff the input has no dual effect. Active learning augments the cost/constraints to induce informative inputs.

Benefits. The correct formulation of control under uncertainty; the conceptual basis for exploration–exploitation in control.

Comparison with ContEst. Dual control is the *objective* CONTEST’s second stage adopts: J_{stoch} in (1) is the dual-control value, and expressing it through the information state is what lets the sensing operator enter the design. The distinction is *offline hardware vs. online policy*: dual/adaptive MPC adapts the *inputs* for a *fixed* design θ ; CONTEST optimizes the *design* θ against the (approximated) closed-loop information dynamics. The two are complementary—the receding-horizon controller keeps running while CONTEST re-provisions parameters. CONTEST is honest about the gap: its tractable eKF surrogate is *caution-aware* (it prices $\text{tr}(Q\Sigma_{t|t})$) but, after linearizing the information dynamics, certainty-equivalent-in-mean—it captures caution but not active probing, in the Bar-Shalom–Tse sense.

8 Differentiable optimization layers

Amos & Kolter, “OptNet: Differentiable Optimization as a Layer,” ICML 2017 [16]; *Agrawal et al.*, “Differentiable Convex Optimization Layers,” NeurIPS 2019 [17].

Method. Embed an optimization problem as a differentiable layer and back-propagate through its *solution* (the minimizer). OptNet differentiates a QP by implicitly differentiating its KKT conditions; Agrawal et al. generalize to any convex program by canonicalizing to a cone program and differentiating the cone-solution (residual) map.

Mathematics. Implicit function theorem applied to the KKT/cone optimality system: $\partial z^*/\partial \theta$ is obtained by solving an $m \times m$ linear system (the differentiated KKT matrix), where m is the number of primal–dual unknowns; cost $O(m^3)$ per direction. This yields the sensitivity of the *argmin*.

Benefits. Fully general end-to-end differentiation of embedded optimizers; ideal when the downstream loss depends on the *minimizer* $z^*(\theta)$ itself (e.g. a learned policy or control action).

Comparison with ContEst. This is the mechanism CONTEST deliberately *avoids*. CONTEST needs the gradient of the inner *value* $V(\theta)$, not of the minimizer; the envelope theorem supplies it

directly from the already-returned dual—no $\partial z^*/\partial\theta$, no KKT-system solve. The complexity gap is quantitative: implicit differentiation solves an $m \times m$ system with $m = O(r_x^2)$ for the covariance SDPs, i.e. $O(r_x^6)$ per directional derivative, whereas the envelope contraction is $O(r_x^2 r_\theta)$ for the *whole* gradient. CONTEST uses these tools’ theoretical backdrop (sensitivity analysis) but not their computation.

9 The engine: envelope theorem and perturbation analysis

Milgrom & Segal, “Envelope Theorems for Arbitrary Choice Sets,” *Econometrica* 2002 [18]; *Bonnans & Shapiro*, *Perturbation Analysis of Optimization Problems*, Springer 2000 [19].

Method. These are *tools*, not competitors. The envelope theorem gives the derivative of an optimal-value function with respect to parameters as the partial derivative of the Lagrangian at the optimizer, holding the optimizer fixed. Bonnans & Shapiro give the rigorous conditions (Slater, unique primal, nondegeneracy/strict complementarity) and the directional-derivative formulas for optimal values of (conic/semidefinite) programs under data perturbations.

Mathematics. For $V(\theta) = \min_z \{\phi(z, \theta) : g = 0, G \in \mathcal{K}\}$ with optimal triple (z^*, μ^*, S^*) , $\nabla_\theta V = \partial_\theta \phi + \langle \mu^*, \partial_\theta g \rangle + \langle S^*, \partial_\theta G \rangle$; the directional derivative equals a minimization over the dual optimal set, collapsing to a gradient when that set is a singleton (or, as ContEst shows, when the value is C^1).

Benefits. Exact value sensitivity computed from the optimizer alone, at negligible cost beyond the primal–dual solve.

Comparison with ContEst. CONTEST’s core contribution is recognizing that the control/estimation *co-design coupling derivative is exactly an envelope gradient*, readable from the inner SDP, Riccati, or MPC duals. Where the differentiable-optimization layers of Section 8 differentiate the minimizer, CONTEST invokes these results to differentiate only the value—turning the coupling derivative from an expensive sweep into a dual contraction.

10 Adjacent threads used for positioning

Gevers, “Identification for Control / Least-Costly Experiment Design,” *EJC* 2005 [20]. *Method.* Design the identification experiment of minimum cost subject to a closed-loop performance requirement. **Comparison with ContEst.** This is the identification-community sibling of designing the noise level $V(\theta)$ against closed-loop cost; CONTEST does the analogous trade *inside* the control loop rather than as a separate identification stage.

Siarni & Jadbabaie, “A Separation Theorem for Joint Sensor and Actuator Scheduling,” *Automatica* 2020 [21]. *Method.* Establishes a separation principle for joint sensor/actuator *scheduling* with guaranteed performance bounds. **Comparison with ContEst.** Like Tzoumas 2021 it relies on a design-level separation and a discrete (scheduling) decision; CONTEST keeps separation only inside the inner solve and treats continuous, coupled design axes.

Zardini, Censi, Frazzoli, “Co-Design of Autonomous Systems,” *ECC* 2021 [22]. *Method.* A compositional (monotone-theory) framework for co-design from hardware selection to control synthesis. **Comparison with ContEst.** Complementary in scope: it composes design problems at a systems level, whereas CONTEST supplies the continuous, gradient-based inner co-design of actuation and estimation that such a framework could call as a component.

Ramadan & Anitescu, “eKF–Koopman for Tractable Stochastic Optimal Control,” *IEEE L-CSS 2024* [23]; Anderson & Moore, *Optimal Filtering*, 1979 [24]. *Method.* The eKF and Koopman-lifted information-state machinery underlying CONTEST’s nonlinear regime. **Comparison with ContEst.** These provide the tractable information state (and a sharper lift than the plain eKF) that CONTEST’s nonlinear surrogate is built on; the eKF–Koopman lift narrows the moment-closure gap of the surrogate.

11 Summary

Table 1 places each reference class against the design axes and the gradient mechanism. The recurring pattern: prior work designs *one* of plant/actuation, sensing, or the estimator—and typically differentiates by finite differences, greedy/relaxation, alternating LMIs, or implicit KKT differentiation. CONTEST closes all three loops in one stochastic objective and obtains the design gradient exactly from inner duals via the envelope theorem.

Table 1: Reference classes vs. design axes and gradient mechanism. ✓ present, – absent, ~ partial/menu-only.

reference class	plant/ actuation	sensing	estimator itself	closed loop	gradient / derivative
CCD / MDO [2, 3, 4]	✓	–	–	✓	finite diff. / adjoint
LMI structural / info-arch. [5, 6]	✓	~	–	✓	alternating LMIs
Sensor selection [7, 8, 9]	–	✓	–	–	greedy / relaxation
Sparse sensing [10, 11, 12]	~	✓	–	~	ℓ_1 / ADMM
LQG sensing co-design [1, 21]	–	✓ (menu)	–	✓	greedy + separation
Dual / adaptive MPC [13, 14, 15]	–	–	–	✓ (online)	n/a (input policy)
Diff. opt. layers [16, 17]	–	–	–	✓	implicit KKT diff.
ContEst	✓	✓ (cont.+ ℓ_1)	✓	✓	exact envelope (dual)

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